EXPERIMENT NO. 13 – CYCLOMATIC COMPLEXITY OF A FLOW GRAPH

AIM

To find the cyclomatic complexity of a flow graph having 17 edges, 13 nodes and 5 predicate nodes, and to verify the result using the alternate formula.

PROCEDURE

1. Open RAPTOR and create a new flowchart.
2. Insert three Input symbols to read the number of edges, nodes and predicate nodes.
3. Insert an Assignment symbol and calculate V1 ← edges - nodes + 2.
4. Insert another Assignment symbol and calculate V2 ← predicates + 1.
5. Insert two Output symbols to display V1 and V2.
6. Run the flowchart using edges = 17, nodes = 13 and predicates = 5.
7. Verify that both formulas produce the same cyclomatic complexity.

OUTPUT

Given: E = 17, N = 13, P = 5

V(G) = E − N + 2 = 17 − 13 + 2 = 6

V(G) = P + 1 = 5 + 1 = 6

Both methods produce the output 6.

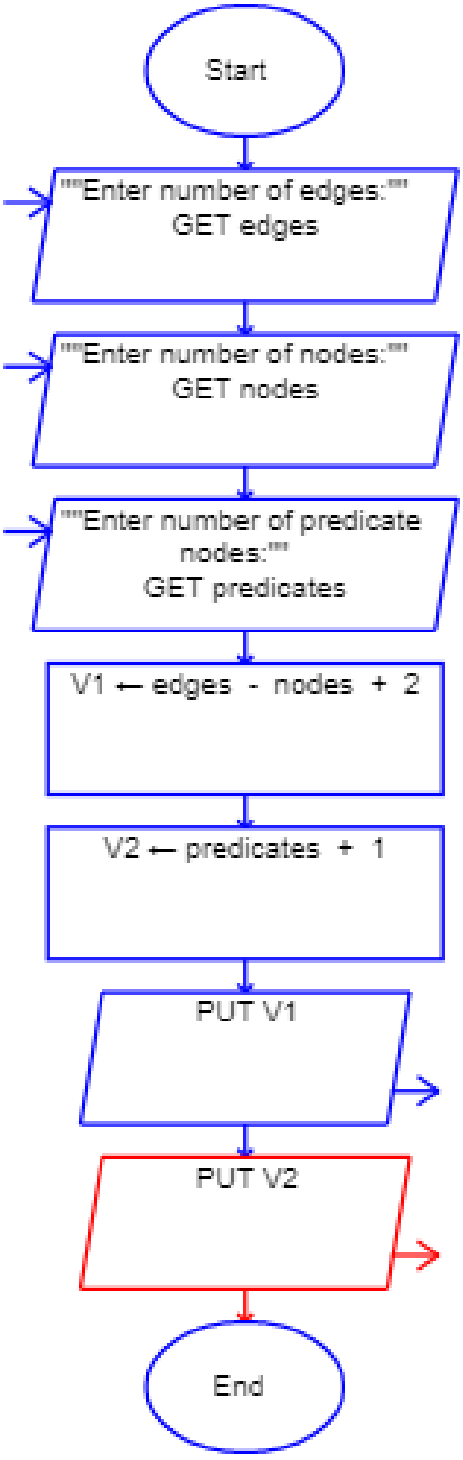


Figure 1 – RAPTOR flowchart

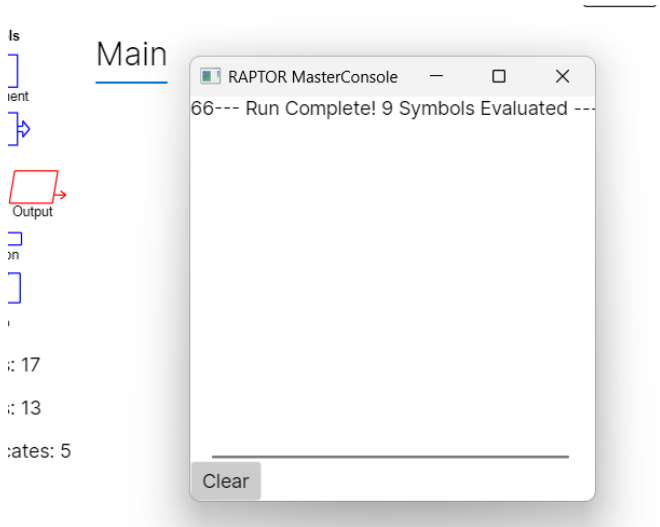


Figure 2 – RAPTOR execution output

RESULT

Thus, the cyclomatic complexity of the given flow graph was calculated as V(G) = 6 using both formulae, indicating six independent paths.